

Creating Roles

Create an Ansible role from an existing Ansible Playbook on workstation, and then create a playbook to apply the role to serverb.lab.example.com and serverc.lab.example.com.

Outcomes

- Create a role from an existing playbook.
- Create a playbook to apply the role to managed hosts.

Specifications

- The review-cr4 directory contains your Ansible project for this activity.
- Convert the ansible-httpd.yml playbook in the project directory into a new Ansible role named ansible-httpd. The new role must be created in the /home/student/review-cr4/roles/ansible-httpd directory.
- Copy any variables, tasks, templates, files, and handlers that were used in or by the playbook into the appropriate files or directories in the new role. Copy the playbook variables to the roles/ansible-httpd/defaults/main.yml file.
- Update the meta/main.yml file in the role with the following content:

Variable	Value
author	Red Hat Training
description	example role for RH294
company	Red Hat
license	BSD

- Edit the roles/ansible-httpd/README.md file so that it provides the following information about the role:

```
ansible-httpd
=====
Example ansible-httpd role from "Red Hat Enterprise Linux Automation with Ansible" (RH294)

Role Variables
-----

* `web_package`: the RPM package
* `web_service`: the systemd service
* `web_config_file`: the path to the main configuration file
* `web_root`: the path to an index.html file
* `web_fw_service`: the name of a firewall service

Dependencies
-----

None.

Example Playbook
-----

- hosts: servers
  roles:
    - ansible-httpd

License
-----

BSD

Author Information
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Red Hat (training@redhat.com)
```

- Remove any unused directories and files within the role.
- In the project directory, write a site.yml playbook that runs the new ansible-httpd role on the managed hosts in the webdev inventory group.
- Run the site.yml playbook.

1. Use the ansible-httpd.yml playbook to create a new Ansible role named ansible-httpd.
 - 1.1. Change into the /home/student/review-cr4 directory.
 - 1.2. Create the roles subdirectory.
 - 1.3. Using the ansible-galaxy command, create the directory structure for the new ansible-httpd role in the roles subdirectory.
 - 1.4. Use the tree command to verify the directory structure created for the new role.

```
[student@workstation review-cr3]$ cd ~/review-cr4
[student@workstation review-cr4]$ pwd
/home/student/review-cr4
[student@workstation review-cr4]$ mkdir -v roles
mkdir: created directory 'roles'
[student@workstation review-cr4]$ cd roles
[student@workstation roles]$ ansible-galaxy role init ansible-httpd
- Role ansible-httpd was created successfully
[student@workstation roles]$ ls
ansible-httpd
[student@workstation roles]$ cd .
[student@workstation roles]$ cd ..
```

```
[student@workstation review-cr4]$ tree roles
roles
```

```
├── ansible-httpd
│   ├── defaults
│   │   └── main.yml
│   ├── files
│   ├── handlers
│   │   └── main.yml
│   ├── meta
│   │   └── main.yml
│   ├── README.md
│   ├── tasks
│   │   └── main.yml
│   ├── templates
│   ├── tests
│   │   ├── inventory
│   │   └── test.yml
│   └── vars
│       └── main.yml
```

9 directories, 8 files

2. Copy any variables, tasks, templates, files, and handlers into the appropriate files inside the new role.
 - 2.1. Copy the variables from the ansible-httpd.yml file into the roles/ansible-httpd/defaults/main.yml file. The roles/ansible-httpd/defaults/main.yml file should contain the following content:

```
---
# defaults file for ansible-httpd
web_package: httpd
web_service: httpd
web_config_file: /etc/httpd/conf/httpd.conf
web_root: /var/www/html/index.html
web_fw_service: http
```

- 2.2. Copy the httpd configuration file template from templates/httpd.conf.j2 into the roles/ansible-httpd/templates/ directory.

```
[student@workstation review-cr4]$ cp -v templates/httpd.conf.j2 roles/ansible-httpd/templates/  
'templates/httpd.conf.j2' -> 'roles/ansible-httpd/templates/httpd.conf.j2'
```

- 2.3. Copy the tasks from the ansible-httpd.yml file into the roles/ansible-httpd/tasks/main.yml file. The roles/ansible-httpd/tasks/main.yml file should contain the following content:

tasks file for ansible-httpd

- name: Packages are installed

ansible.builtin.dnf:

name: "{{ web_package }}"

state: present

- name: Ensure service is started

ansible.builtin.service:

name: "{{ web_service }}"

state: started

enabled: true

- name: Deploy configuration file

ansible.builtin.template:

src: templates/httpd.conf.j2

dest: "{{ web_config_file }}"

owner: root

group: root

mode: '0644'

setype: httpd_config_t

notify: Restart httpd

- name: Deploy index.html file

ansible.builtin.copy:

src: files/index.html

```

    dest: "{{ web_root }}"

    owner: root

    group: root

    mode: '0644'

- name: Web port is open

  ansible.posix.firewalld:

    service: "{{ web_fw_service }}"

    permanent: true

    state: enabled

    immediate: true

  name: "{{ web_service }}"
  state: started
  enabled: true

- name: Deploy configuration file
  ansible.builtin.template:
    src: templates/httpd.conf.j2
    dest: "{{ web_config_file }}"
    owner: root
    group: root
    mode: '0644'
    setype: httpd_config_t
    notify: Restart httpd

- name: Deploy index.html file
  ansible.builtin.copy:
    src: files/index.html
    dest: "{{ web_root }}"
    owner: root
    group: root
    mode: '0644'

- name: Web port is open
  ansible.posix.firewalld:
    service: "{{ web_fw_service }}"
    permanent: true
    state: enabled
    immediate: true

```

2.4. Copy the files/index.html file into the roles/ansible-httpd/files/ directory.

```

[student@workstation review-cr4]$ cp -v files/index.html roles/ansible-httpd/files/
'files/index.html' -> 'roles/ansible-httpd/files/index.html'

```

2.5. Copy the handlers from the ansible-httpd.yml file into the roles/ansible-httpd/handlers/main.yml file. The roles/ansible-httpd/handlers/main.yml file should contain the following content:

```

[student@workstation review-cr4]$ vi roles/ansible-httpd/handlers/main.yml
[student@workstation review-cr4]$ cat roles/ansible-httpd/handlers/main.yml
---
# handlers file for ansible-httpd
- name: Restart httpd
  ansible.builtin.service:
    name: "{{ web_service }}"
    state: restarted

```

3. Update the roles/ansible-httpd/meta/main.yml file in the role according to the specifications.
 - 3.1. Change the value of the author entry to Red Hat Training.
 - 3.2. Change the value of the description entry to example role for RH294.
 - 3.3. Change the value of the company entry to Red Hat.
 - 3.4. Change the value of the license entry to BSD.
-

```
galaxy_info:
  author: Red Hat Training
  description: example role for RH294
  company: Red Hat
  license: BSD

# If the issue tracker for your role is not on github, uncomment the
# next line and provide a value
# issue_tracker_url: http://example.com/issue/tracker

# Choose a valid license ID from https://spdx.org - some suggested licenses:
# - BSD-3-Clause (default)
# - MIT
# - GPL-2.0-or-later
# - GPL-3.0-only
# - Apache-2.0
# - CC-BY-4.0
license: license (GPL-2.0-or-later, MIT, etc)

min_ansible_version: 2.1

# If this a Container Enabled role, provide the minimum Ansible Container version.
# min_ansible_container_version:

#
# Provide a list of supported platforms, and for each platform a list of versions.
# If you don't wish to enumerate all versions for a particular platform, use 'all'.
# To view available platforms and versions (or releases), visit:
```

4. Edit the roles/ansible-httpd/README.md file so that it provides pertinent information regarding the role. The file should consist of the following content:

```
* `web_package`: the RPM package
* `web_service`: the systemd service
* `web_config_file`: the path to the main configuration file
* `web_root`: the path to an index.html file
* `web_fw_service`: the name of a firewall service
```

Dependencies

None.

Example Playbook

```
- hosts: servers
  roles:
    - ansible-httpd
```

License

BSD

Author Information

Red Hat (training@redhat.com)

5. Remove the unused directories from the new role.
 - 5.1. Remove the roles/ansible-httpd/vars/ directory.
 - 5.2. Remove the roles/ansible-httpd/tests/ directory.

```
[student@workstation review-cr4]$ rm -rfv roles/ansible-httpd/vars/
removed 'roles/ansible-httpd/vars/main.yml'
removed directory 'roles/ansible-httpd/vars/'
[student@workstation review-cr4]$ rm -rfv roles/ansible-httpd/tests/
removed 'roles/ansible-httpd/tests/inventory'
removed 'roles/ansible-httpd/tests/test.yml'
removed directory 'roles/ansible-httpd/tests/'
[student@workstation review-cr4]$ vi site.yml
[student@workstation review-cr4]$ cat site.yml
```

6. In the project directory, write a site.yml playbook that runs the new ansible-httpd role on the managed hosts in the webdev inventory group. The site.yml playbook should contain content similar to the following example:

```
[student@workstation review-cr4]$ vi site.yml
[student@workstation review-cr4]$ cat site.yml
---
- name: Apply the ansible-httpd role
  hosts: webdev

  roles:
    - ansible-httpd
```

7. Run the site.yml playbook.

ansible-navigator run -m stdout site.yml

```
[student@workstation review-cr4]$ ansible-navigator run -m stdout site.yml
[WARNING]: While constructing a mapping from /home/student/review-cr4/roles/ansible-httpd/meta/main.yml, line 2, column 3, found a duplicate dict key (license). Using last defined value only.
```

```
PLAY [Apply the ansible-httpd role] *****
```

```
TASK [Gathering Facts] *****
```

```
ok: [serverc.lab.example.com]
```

```
ok: [serverb.lab.example.com]
```

```
TASK [ansible-httpd : Packages are installed] *****
```

```
changed: [serverc.lab.example.com]
```

```
changed: [serverb.lab.example.com]
```

```
TASK [ansible-httpd : Ensure service is started] *****
```

```
changed: [serverb.lab.example.com]
```

```
changed: [serverc.lab.example.com]
```

```
TASK [ansible-httpd : Deploy configuration file] *****
```

```
changed: [serverb.lab.example.com]
```

```
changed: [serverc.lab.example.com]
```

```
TASK [ansible-httpd : Deploy index.html file] *****
```

```
changed: [serverb.lab.example.com]
```

```
changed: [serverc.lab.example.com]
```

```
TASK [ansible-httpd : Web port is open] *****
```

```
changed: [serverc.lab.example.com]
```

```
changed: [serverb.lab.example.com]
```

```
RUNNING HANDLER [ansible-httpd : Restart httpd] *****
```

```
changed: [serverb.lab.example.com]
```

```
changed: [serverc.lab.example.com]
```

```
PLAY RECAP *****
```

serverb.lab.example.com	: ok=7	changed=6	unreachable=0	failed=0	skipped=0	rescued=0	ignored=0
serverc.lab.example.com	: ok=7	changed=6	unreachable=0	failed=0	skipped=0	rescued=0	ignored=0